Keeping remote compressors running

Field service with hybrid AI agents, seen by the supervisor and by the technician who does the repair

hybrid-ai · November 2026

Supervisor Technician

The situation: a spread-out compressor fleet

30

compressors

industrial: screw, reciprocating, centrifugal, diesel, scroll

12

remote sites

valley, desert and foothill locations

6

depots

each with an edge server; 4 with workshops

12

technicians

in service vans, different skills

Every fault starts the same chain of decisions

What's wrong? → **Who goes?** → **On site or workshop?** → **Are the parts in the van?** → **How do I fix it?** → **Record it**

The problem: half the compressors are at sites with patchy or no coverage, and about 1 in 5 past repairs needed a return visit.

The challenges

1

No signal where the work is

Dead zones and wildfire power shutoffs cut the link for hours or days.

2

Live data is heavy

Vibration, temperature and pressure can't all stream over one bar of LTE.

3

Knowledge is scattered

Manuals, past repairs and part lists live in different places.

4

The cloud AI can fail too

Model outages happen even when the network is fine.

5

Safety can't wait

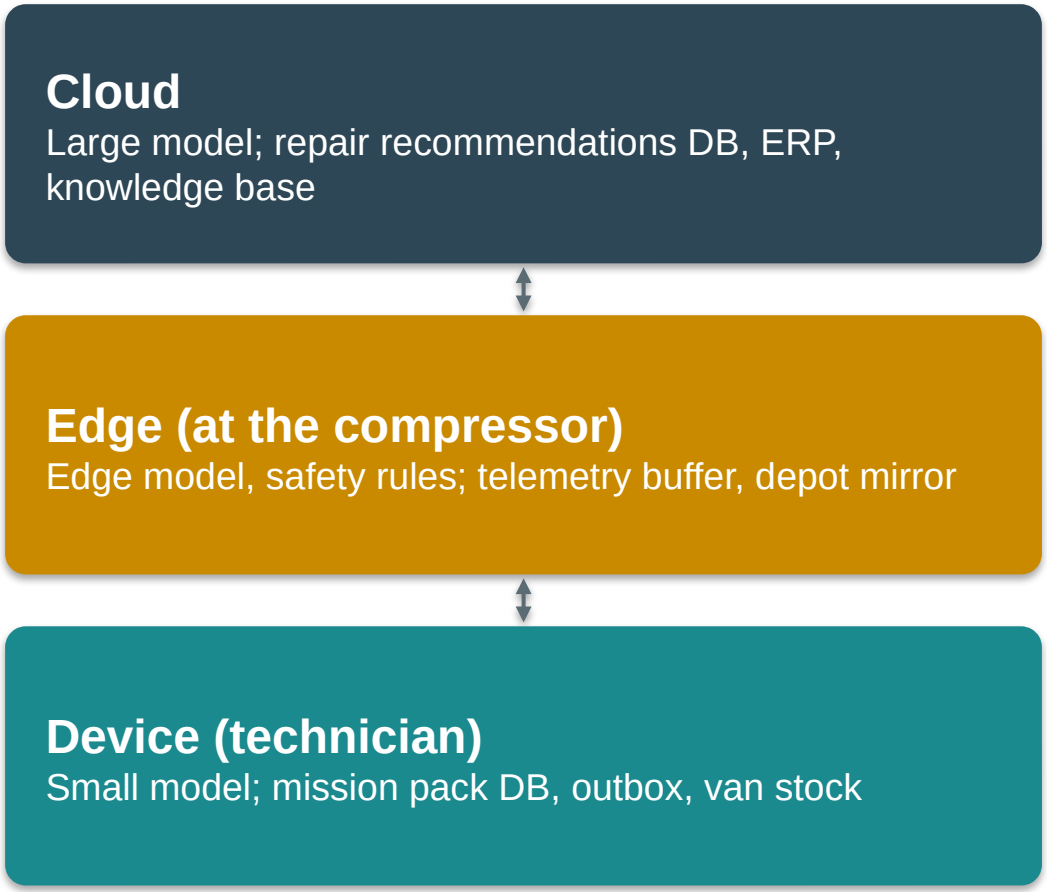
Lockout, shutdowns and approvals must work offline and never use stale data.

6

Updates get lost

Notes taken offline are retyped later, or never reach the office.

Solution: agents move between three places



1

A router picks where each request runs

Cloud when Online, the edge model when the link is Limited or gone, the device when the edge is dark too.

2

Every answer is labeled

Where it ran, why, how fresh it is, and what it cost in time and tokens.

3

A trained model and an AI agent

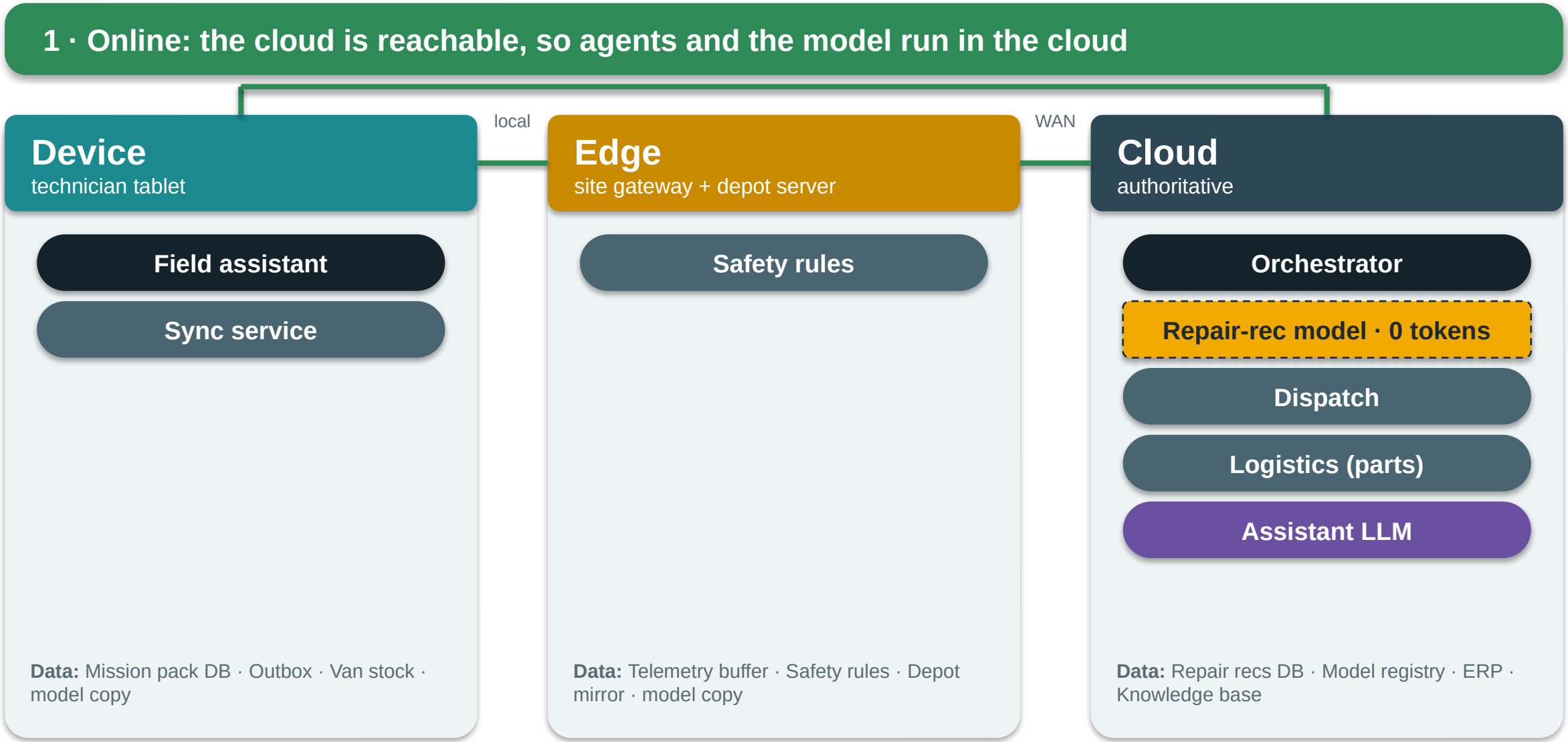
A 9 KB model ranks causes in every place; a Claude agent searches the service manuals and writes cited repair steps (retrieval only when offline).

4

Nothing is lost offline

Updates queue in priority order and replay safely once, when the link returns.

Where agents run as the network changes



Starting state (Online). The PowerPoint version animates all five stages. · square chip = trained model, purple = LLM, dark = agent, grey = tool/service

Supervisor's view: decide fast, see everything

Supervisor

1

Review

Each work order arrives with a ranked diagnosis, best technicians, repair location and parts.

2

Approve

Rules approve routine work; urgent dispatch and parts over $1,000 come to the supervisor.

3

Track

A live panel shows every request: where it ran, why, and the time and tokens it cost.

4

Close

Field updates appear as they sync; the repair outcome lands in the database.

What's different

No guessing about data age: every answer says which place answered and when

Approvals by rule, so only the decisions that need a person reach one

Knows exactly which technician updates are still in transit

Technician's view: one repair, through a dead zone

Technician

1	2	3	4	5	6	7
ONLINE	LIMITED	OFFLINE	OFFLINE	OFFLINE	LIMITED	ONLINE
Depot	**Highway**	**On site**	**Repair**	**Close out**	**Driving out**	**Back online**
Plan copied into the mission pack DB; cloud model answers	Weak signal: the edge model answers; cloud only in windows	Live readings over the local link; lockout steps queued	Gateway dark: the device model takes over; plan continues	Test run, findings and photo go to the outbox	Safety records and job updates replay first	Photo uploads; the cloud DB shows the job complete

What's different: the plan, manuals, past fixes, live readings and van stock are on the device. The assistant still answers offline, and every step the technician ticks reaches the database later, exactly once.

When the network or the cloud fails

Mode or event	Agents run on	Supervisor sees	Technician can still
Online	Cloud (+ edge for live data)	Everything live	Everything, with the large cloud model
Limited	Edge and device; cloud in windows	Field updates arrive in bursts	All work; photos wait for Online
Offline	Edge and device only	Last known state, marked stale	Plan, manuals, readings, lockout, outbox
Cloud AI outage	Edge model; cloud data stays live	Live data, slower answers	Same as normal, slower answers
Sensors lost	Device (edge is dark)	No live data, clearly flagged	Guided inspection instead of telemetry

Implementation architecture

Device :8803 technician laptop or tablet

- Router: per-mode policy and deadlines
- Link monitor: probes decide the mode
- Mission pack and outbox (SQLite)
- Repair model and manual index copies
- Web page: live.html

Edge :8802 depot server

- MCP server: telemetry, diagnosis, parts
- Live telemetry files (JSONL)
- Repair model and manual index copies
- Pushes features to the cloud

Cloud :8801 central services

- MCP server: plans, parts, steps, outcomes
- Repair recommendations DB (SQLite)
- Manual agent: Claude Opus 5.5 + RAG
- Fleet, technicians, parts stock

netsim :8810-8813 a TCP proxy on every link that delays, flaps or cuts it · all calls: MCP over streamable HTTP

One Python package: src/hybrid_ai/

`contracts`
JSON schemas for every message

`sim`
fleet, telemetry, network modes

`models`
repair model, 9 KB, no tokens

`rag`
BM25 index over the manuals

`agents`
the Claude manual agent

`live`
the four processes and netsim

`demo`
dashboard server and repair DB

Dashboard :8765
simulated supervisor and technician views, JSON API over the repair DB

Deployment
systemd user or system units · Ubuntu tarball · hosted copy on claude.ai

Stack
Python 3 · mcp 2.3 · uvicorn · SQLite · anthropic SDK · Leaflet